



TABLE I
STANDARD SIZES - RECOMMENDED FOR NEW DESIGN AND ENGINEERING

DASH NUMBER		TUBE O.D.	A PIPE THREAD	B STRAIGHT THREAD	C HEX FLAT	D DIA	D_1 DIA	L	S
ASSY	*BODY	NOM	NPTF	CLASS 2A	NOM	DRILL	DRILL	±.02	MAX
1	31	1/8	1/8	5/16-24UNF	7/16	0.093	0.188	1.04	0.49
2	32	3/16	1/8	3/8 -24UNF	7/16	0.125	0.188	1.09	0.49
3	33	1/4	1/8	7/16-20UNF	1/2	0.203	0.188	1.12	0.55
4	34	5/16	1/8	1/2 -20UNF	9/16	0.234	0.188	1.12	0.55
7	37	3/8	1/4	9/16-18UNF	5/8	0.281	0.281	1.34	
9	39	1/2	3/8	3/4 -16UNF	13/16	0.422	0.406	1.44	0.67
(F) 12	42	5/8	1/2	7/8 -14UNF	7/8	0.500	0.531	1.75	0.91
14	44	3/4	3/4	1-1/16-12UN	1-1/8	0.656	0.719	1.88	0.94
15	45	7/8	3/4	1-3/16-12UN	1-1/4	0.718	0.719	1.88	
18	48	1	1	1-5/16-12UN	1-3/8	0.875	0.938	2.07	1.13
20	50	1-1/4	1-1/4	1-5/8 -12UN	1-11/16	1.093	1.250	2.18	1.20
21	51	1-1/2	1-1/2	1-7/8 -12UN	2	1.344	1.500	2.28	1.27
22	52	2	2	2-1/2 -12UN	2-3/8	1.813	1.938	2.46	1.37

TABLE II
NONSTANDARD SIZES - NOT RECOMMENDED FOR NEW DESIGN AND ENGINEERING

DASH NUMBER		TUBE O.D.	A PIPE THREAD	B STRAIGHT THREAD	C HEX FLAT	D DIA	D_1 DIA	L	S
ASSY	*BODY	NOM	NPTF	CLASS 2A	NOM	DRILL	DRILL	±.02	MAX
5	35	1/4	1/4	7/16-20UNF	9/16	0.203	.281	1.32	0.69
6	36	5/16	1/4	1/2 -20UNF	9/16	0.234	.281	1.32	0.69
8	38	1/2	1/4	3/4 -16UNF	13/16	0.422	.281	1.44	0.67
(F) 10	40	5/8	3/8	7/8 -14UNF	7/8	0.500	.406	1.56	0.77
(F) 11	41	3/8	1/2	9/16-18UNF	7/8	0.281	.531	1.60	0.91
13	43	3/4	1/2	1-1/16-12UN	1-1/8	0.656	.531	1.88	0.86
16	46	7/8	1	1-3/16-12UN	1-3/8	0.718	.938	2.07	1.13
17	47	1	3/4	1-5/16-12UN	1-3/8	0.875	.719	1.88	0.86
19	49	1-1/4	1	1-5/8 -12UN	1-11/16	1.093	.938	2.15	0.90

FOR NOTES SEE SHEET 2.

(E) Denotes change

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(E) APPROVED 29 DEC 1967 REVISED (A) 30 DEC 1970 (B) 22 NOV 1974 (C) 13 May 77 (D) 17 June 1982 (E) 30 July 1987

P.A. AR	TITLE	MILITARY STANDARD
Other Coor 99	ADAPTER, STRAIGHT, PIPE TO TUBE, MALE PIPE END, FLARELESS TYPE, HYDRAULIC	MS51819
PROCUREMENT SPECIFICATION MIL-F-18866	SUPERSEDES: CNMX1 in part, and 14 in part	SHEET 1 OF 2

DD FORM 672-1 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE 4730-0885

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NOTES:

1. MATERIAL: Steel, carbon in accordance with procurement specification.
   Steel, corrosion-resisting in accordance with procurement specification.

2. PROTECTIVE COATING: Cadmium plated in accordance with procurement specification.
   Zinc plated in accordance with procurement specification.
   Phosphate coating in accordance with procurement specification.
   Passivation treatment in accordance with procurement specification.

3. DIMENSIONS AND TOLERANCES: Dimensions and tolerances not shown shall be in accordance with the SAE Standard for Flareless Tube Fittings (SAE J514).

4. THREADS: Threads shall be in accordance with Screw-Thread Standards for Federal Services, FED-STD-H28.

(E) 5. PART NUMBER: The MS part number consists of the MS number, plus the dash number and (for complete assembly) the applicable MS51825 sleeve style designator.

   A. For complete assembly (with nut and sleeve), cadmium plated, use the MS number, plus the assembly dash number and the sleeve style designator. Example: MS51819-1A or MS51819-18.

   B. For the body only, cadmium plated, use the MS number and the body dash number. Example: MS51819-31.

   C. For zinc coating add "Z" and (for complete assembly) the sleeve style designator to the dash number. Example: MS51819-1ZA or MS51819-1ZB or MS51819-31Z.

   D. For phosphate coating add "P" and (for complete assembly) the sleeve style designator to the dash number. Example: MS51819-1PA or MS51819-1PB or MS51819-31P.

   E. For corrosion-resisting steel add "SS" and (for complete assembly) the sleeve style designator to the dash number. Example: MS51819-1SSA or MS51819-1SSB or MS51819-31SS.

6. For design feature purposes, this standard takes precedence over procurement documents referenced herein.

7. Reference documents shall be of the issue in effect on date of invitations for bid.

8. The illustration is for identification and is not intended to restrict designs or shapes not dimensioned.

9. At manufacturer's option, through passages may conform with the smaller diameter specified or the appropriate end may be counterbored to the larger diameter for depth S.

* 10. After 22 NOV 1974, bodies will be used for production only and SHALL NOT be stocked for maintenance. Existing stock of bodies should be used until depleted.

11. CAUTION: The following data is furnished for the purpose of new design only and is not intended to imply interchangeability.

(E) MS51819 will replace MS39206 for new design after approval date on this document. Table III below gives reference data for coordination of equivalent tube sizes, in both compression and hydraulic type adapters. Assembly replacement is authorized.

TABLE III (FOR INFORMATION ONLY)
ASSEMBLY EQUIVALENCY

PART NUMBER		PART NUMBER		PART NUMBER	
COMPRESSION	HYDRAULIC	COMPRESSION	HYDRAULIC	COMPRESSION	HYDRAULIC
MS39206-1	MS51819-1	MS39206-8	MS51819-8	MS39206-15	MS51819-15
-2	-2	-9	-9	-16	-16
-3	-3	-10	-10	-17	-17
-4	-4	-11	-11	-18	-18
-5	-5	-12	-12	-19	-19
-6	-6	-13	-13	-20	-20
-7	-7	-14	-14	-21	-21

APPROVED 29 DEC 1987 REVISED (A) 30 DEC 1970 (B) FOR CHANGES SEE SHEET 1 (C) FOR CHANGES SEE SHEET 2. (D) 17 June 1982 (E) 30 July 1987

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Army - AT, CL, ME
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(E)

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P.A. AR Other Cust 99	TITLE ADAPTER, STRAIGHT, PIPE TO TUBE, MALE PIPE END, FLARELESS TYPE, HYDRAULIC	MILITARY STANDARD MS51819
PROCUREMENT SPECIFICATION MIL-F-18866	SUPERSEDES: CNHKI in part, and 14 in part	SHEET 2 OF 2

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